How does this Shiny dashboard work?

From user selections to linked maps, graphs, and tables

INPUTS

County

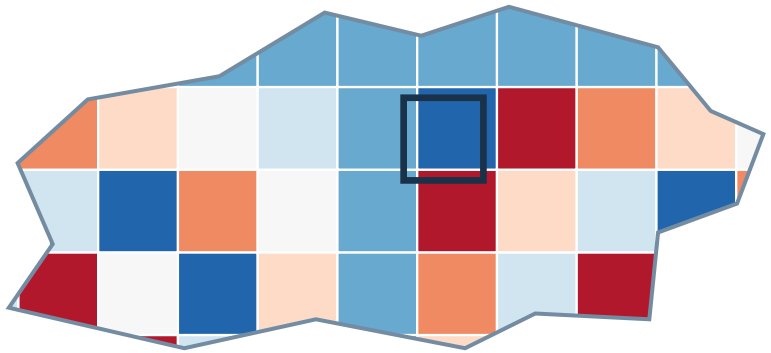
Cuyahoga ▾

Years

1975 1980

Shiny inputs

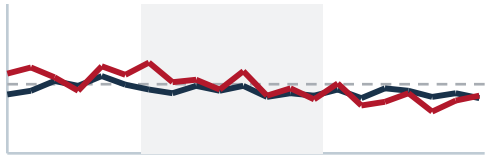
LEAFLET MAP



Selected county

low SIR high SIR

PLOTLY TIME SERIES



DT TABLE

COUNTY	SIR
Example A	1.46
Example B	1.23
Cuyahoga	1.11
Example D	0.91

Two layers: document + application

R Markdown provides the layout; Shiny supplies reactivity

1 DOCUMENT

```
YAML header:
  output: flexdashboard::flex_dashboard
  orientation: columns
  runtime: shiny
```

2 APPLICATION

```
Shiny code:
  selectInput(...)
  reactive({ ... })
  renderLeaflet({ ... })
```

DATA SETUP (runs when app starts)

oh = CSV
county × year; Y, E, GEOID

oh_map = boundaries
geometries keyed by GEOID

ohio_year = statewide SIR
precalculated, once

Input IDs connect the interface to R

The label is for people; the ID is for code

DROPDOWN

```
selectInput(
  "county", "County",
  choices = ..., selected = "Cuyahoga"
)
```

"county" → input$county

County: Cuyahoga

RANGE SLIDER

```
sliderInput(
  "years", "Years",
  min = 1968, max = 1988,
  value = c(1968, 1988)
)
```

"years" → input$years



Shiny creates input automatically. input$years[1] = start year; input$years[2] = end year.

The reactive calculation: observed / expected

Changing the year interval changes county SIR estimates

```
by_county <- reactive({
  oh |> filter(between(year,
             input$years[1], input$years[2])) |>
    group_by(GEOID, county) |>
    summarise(Y=sum(Y), E=sum(E)) |>
    mutate(SIR=Y/E)
})
```

ILLUSTRATIVE COUNTY

Observed (Y)	Expected (E)
720	**600**

SIR = 720 / 600 = 1.20

Key distinction

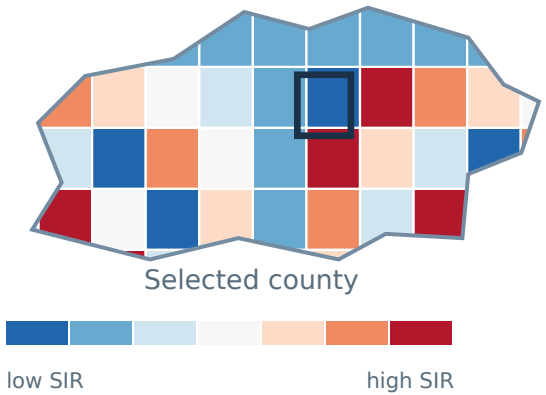
Period SIR = sum(observed cases) / sum(expected cases). **Call the reactive result with by_county()**

Three outputs, three rendering functions

Same data pipeline, different displays

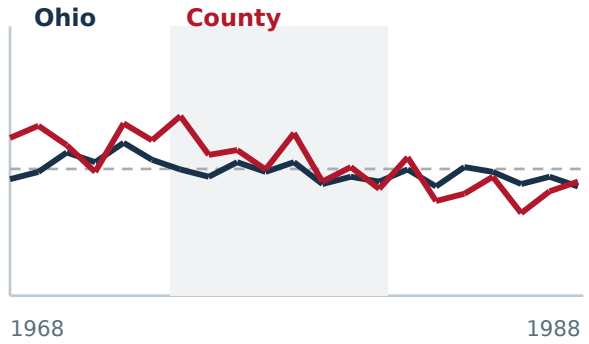
CHOROPLETH

`renderLeaflet({ ... })`



TIME SERIES

`renderPlotly({ ... })`



TABLE

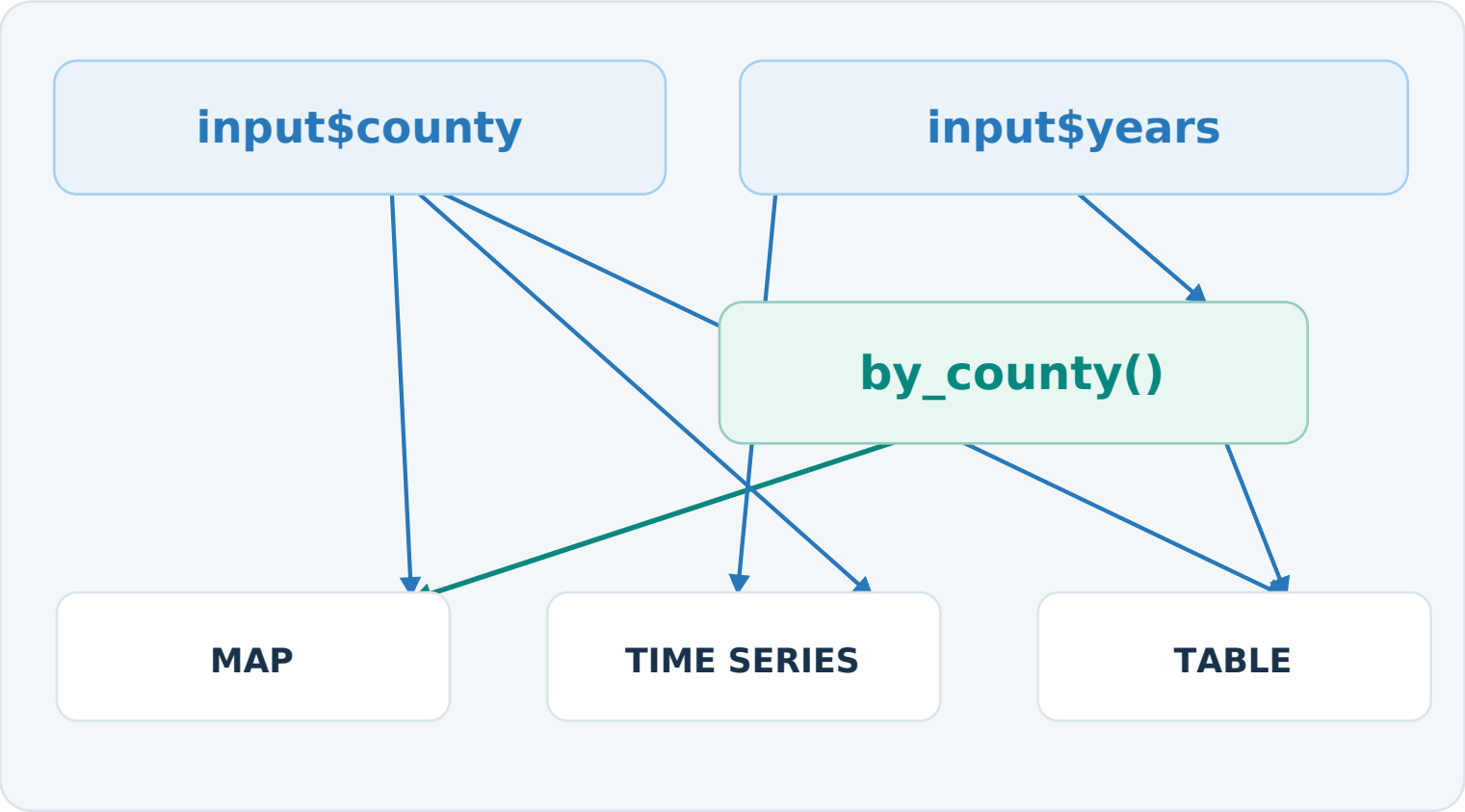
`renderDT({ ... })`

COUNTY	SIR
Example A	1.46
Example B	1.23
Cuyahoga	1.11
Example D	0.91

Map: by_county() + boundaries • **Plot: annual county + Ohio SIR** • **Table: by_county()**

Shiny updates only what depends on an input

This is the reactive dependency graph



CHANGE COUNTY

Map outline, time-series county line, and bold table row change.

by_county() is unchanged.

CHANGE YEARS

by_county() recalculates. The map and table update; the graph shading moves.

Five concepts students should retain

A 15-minute walkthrough, then one question

THE FIVE ESSENTIAL PIECES

```
1  selectInput("county", ...)

2  input$county

3  by_county <- reactive({ ... })

4  by_county()

5  renderLeaflet({ ... })
```

IN CLASS

3 min	Explore the dashboard
3 min	Identify both inputs
4 min	Calculate a period SIR
3 min	Trace the three outputs
2 min	Check understanding

CHECKPOINT QUESTION

If only the county changes, does by_county() recalculate?

One dashboard = inputs → reactive calculations → linked visualizations.